Chapter 17, Problem 6P

Problem

Find the trigonometric Fourier series for

$$f(t) = \begin{cases} 7.5 & 0 < t < \pi \\ 15, & \pi < t < 2\pi \end{cases} \quad \text{and} \quad f(t + 2\pi) = f(t).$$

Step-by-step solution

Step 1 of 5

Consider the function, $f(t)$, with the time period 2π.

$$f(t) = \begin{cases} 7.5, & 0 < t < \pi \\ 15, & \pi < t < 2\pi \end{cases} \quad \text{and} \quad f(t+2\pi) = f(t)$$

Since the function $f(t)$ is repeated for period of 2π. The time period is $T = 2\pi$.

Calculate the fundamental angular frequency.

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2\pi} \\ &= 1 \end{aligned}$$

Write the general form for trigonometric Fourier series of $f(t)$.

$$\begin{aligned} f(t) &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t) \\ &= a_0 + \sum_{n=1}^{\infty} (a_n \cos n(1)t + b_n \sin n(1)t) \\ &= a_0 + \sum_{n=1}^{\infty} (a_n \cos nt + b_n \sin nt) \end{aligned}$$

Step 2 of 5

Calculate the a_0, average value of $f(t)$.

$$\begin{aligned} a_0 &= \frac{1}{T}\int_0^T f(t)\,dt \\ &= \frac{1}{2\pi}\int_0^{2\pi} f(t)\,dt \\ &= \frac{1}{2\pi}\left(\int_0^{\pi} 7.5\,dt + \int_{\pi}^{2\pi} (15)\,dt\right) \\ &= \frac{1}{2\pi}(7.5(\pi - 0) + 15(2\pi - \pi)) \\ &= \frac{1}{2\pi}(22.5\pi) \\ &= 11.25 \end{aligned}$$

Step 3 of 5

Calculate the value of a_n.

$$\begin{aligned} a_n &= \frac{2}{T}\int_0^T f(t)\cos n\omega_0 t dt \\ &= \frac{2}{2\pi}\int_0^{2\pi} f(t)\cos n(1)t dt \\ &= \frac{1}{\pi}\left(\int_0^{\pi} 7.5\cos nt dt + \int_{\pi}^{2\pi}(15\cos nt)dt\right) \end{aligned}$$

Simplify the equation further.

$$\begin{aligned} a_n &= \frac{1}{\pi}\left(7.5\left(\frac{\sin nt}{n}\right)\Bigg|_{t=0}^{\pi} + 15\left(\frac{\sin nt}{n}\right)\Bigg|_{t=\pi}^{2\pi}\right) \\ &= \frac{1}{\pi}\left(7.5(\sin n\pi - \sin n(0)) + \frac{15}{n}(\sin 2n\pi - \sin n\pi)\right) \\ &= \frac{1}{\pi}(0) \\ &= 0 \end{aligned}$$

Step 4 of 5

Calculate the b_n.

$$\begin{aligned} b_n &= \frac{2}{T}\int_0^T f(t)\sin n\omega_0 t dt \\ &= \frac{2}{2\pi}\int_0^{2\pi} f(t)\sin n(1)t dt \\ &= \frac{1}{\pi}\left(\int_0^{\pi} 7.5\sin nt dt + \int_{\pi}^{2\pi}(15\sin nt)dt\right) \end{aligned}$$

Simplify the equation further.

$$\begin{aligned} b_n &= \frac{1}{\pi}\left(7.5\left(\frac{-\cos nt}{n}\right)\Bigg|_{t=0}^{\pi} + 15\left(\frac{-\cos nt}{n}\right)\Bigg|_{t=\pi}^{2\pi}\right) \\ &= \frac{1}{n\pi}\left(-7.5(\cos n\pi - \cos n(0)) - 15(\cos 2n\pi - \cos n\pi)\right) \\ &= \frac{1}{n\pi}\left(-7.5\left((-1)^n - 1\right) - 15\left(1 - (-1)^n\right)\right) \\ &= \frac{1}{n\pi}\left(7.5(-1)^n - 7.5\right) \\ &= \frac{7.5}{n\pi}\left((-1)^n - 1\right) \\ &= \begin{cases} \dfrac{-15}{n\pi}, & \text{if } n \text{ is odd} \\ 0, & \text{if } n \text{ is even} \end{cases} \end{aligned}$$

Step 5 of 5

Thus, the trigonometric Fourier series for the function $f(t)$.

$$\begin{aligned}
f(t) &= a_0 + \sum_{n=1}^{\infty}\left(a_n \cos nt + b_n \sin nt\right) \\
&= 11.25 + \sum_{n=1}^{\infty}\left((0)\cos nt + \left(\begin{cases} \dfrac{-15}{n\pi}, & if\ n \text{ is odd} \\ 0, & if\ n \text{ is even} \end{cases}\right)\sin nt\right) \\
&= 11.25 + \sum_{n=odd}^{\infty}\left(\frac{-15}{n\pi}\sin nt\right) \\
&= 11.25 - \sum_{n=odd}^{\infty}\left(\frac{15}{n\pi}\sin nt\right)
\end{aligned}$$

Thus, the trigonometric Fourier series for the function $f(t)$ is $\boxed{11.25 - \sum_{n=odd}^{\infty}\left(\frac{15}{n\pi}\sin nt\right)}$.